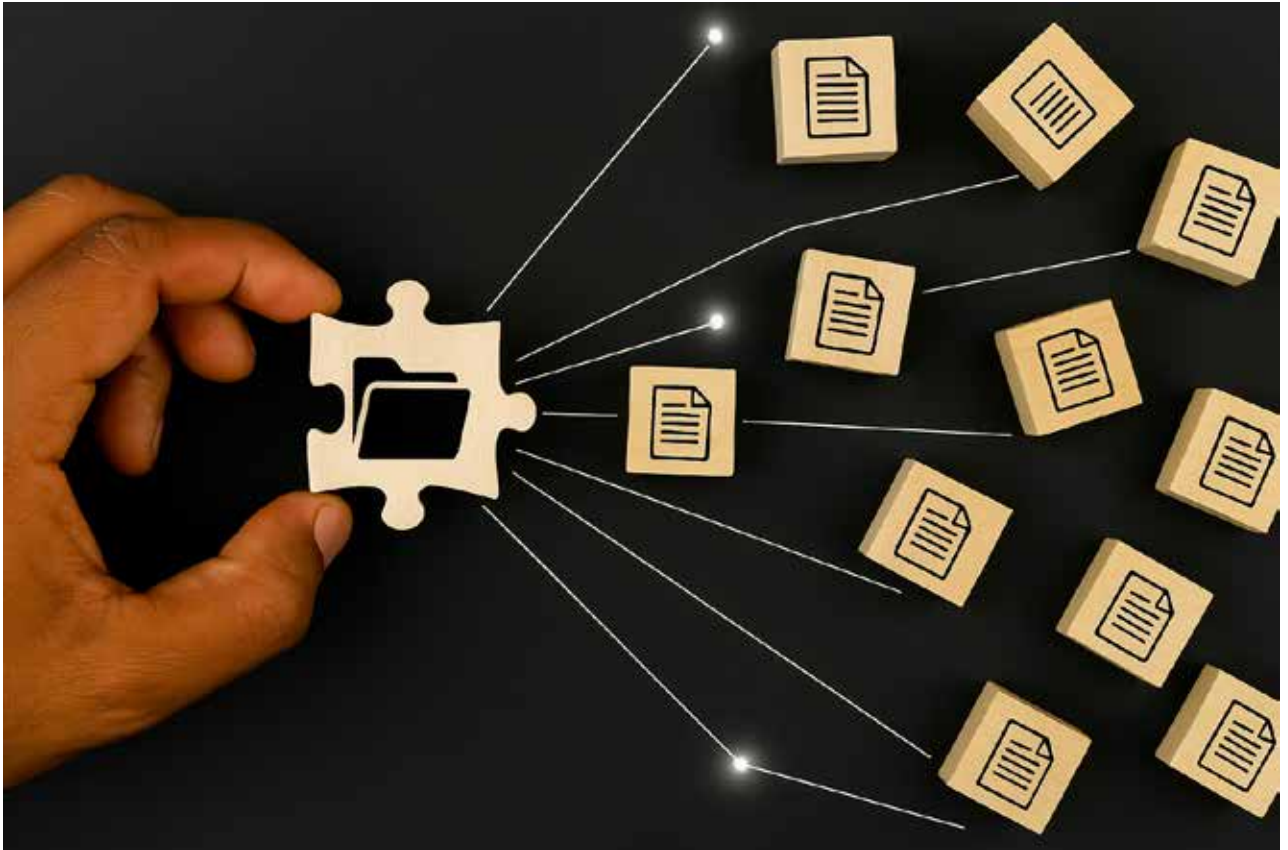


An Introduction to Inodes

Understanding inodes can be invaluable for troubleshooting disk-space issues, optimising filesystem performance, and recovering deleted files.



An inode (short for ‘index node’) is the link between a file name and its physical location on a storage device. In the UNIX and Linux operating systems, inodes are data structures used to hold metadata about files and directories. A filesystem is how an operating system organises and stores files on a storage device. Each filesystem on the computer has its own separate set of inodes. In short, each filesystem mounted on a computer has its inodes, which are stored in a common table. An inode number can be used more than once but not by the same filesystem. The filesystem ID is

paired with the inode number to create a unique identification label.

When a filesystem is created, a fixed number of inodes is created. Typically, approximately 1% of the filesystem disk space is allocated to the inode table. Each file and directory in a filesystem is assigned a unique inode, which is identifiable by an integer known as the inode number. A filesystem is typically separated into inodes and data blocks.

The directory is implemented as a table that maps the filenames to inode numbers. Every file in a directory table is an entry with a filename and inode numbers. All other information about the file is obtained by referencing the

inode number in the inode table. When a file is opened, the operating system uses the inode’s pointers to navigate through the disk and retrieve the file content from its physical location. This process is seamless for users but relies heavily on these ‘directions’ stored in the inodes.

Metadata in inodes

Inodes store various types of metadata (as indicated below) about a file or directory, but do not store the name or actual file data.

File type: The file can be a normal file, a directory, or a symbolic link.